



ARCHITECTURAL
CONSULTING
GROUP
مماراة الاستشارات المعمارية

FIVE STAR HOTEL
PLOT: C104b_02, SECTOR: E16_02
ABU DHABI, UAE



SECTION 22 05 23

PLUMBING – VALVES



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PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings, the general and particular provisions of the Contract, and the sections of the specifications of the Division 01, apply to this Section.

1.2 SUBMISSIONS

- A. Contractor shall produce and submit for approval detailed schedule of all valves covered by this Section indicating type of devices, dimensions, materials of construction, pressure/temperature ratings and attachment methods, for each pipe size and type of service. Temperature ratings specified are for continuous operation.
- B. Contractor shall submit to the ENGINEER a Statement from manufacturer on his letterhead that the product furnished meets the BS specification contained in this Section.

1.3 QUALITY CONTROL AND ASSURANCE

- A. In accordance with the Conditions of Contract, and as specified herein.
- B. All gate, globe and angle valves shall be designed for repacking under pressure when fully opened and shall be equipped with packing suitable for intended service. When valve is fully opened, back seat shall protect the packing and stem threads from fluid.
- C. Bronze gate, globe, angle and check valves shall conform to BS 5154.
- D. Pressure containing parts of valves shall be of material conforming to BS 5150 PN16 & PN25.
- E. Gate valves shall conform to BS 5150.
- F. Valve stems shall be copper silicon alloy conforming to BS 2874.
- G. Wheel handles shall be non-heating style cast from malleable iron or commercial aluminum.
- H. Check valves shall operate equally well in horizontal or vertical positions.
- I. Double regulating valves shall conform to the requirements of BS 7350.
- J. All valves shall be full line size.
- K. All valves shall bear third party quality assurance stamp.



1.4 OPERATION AND MAINTENANCE MANUAL AND TRAINING OF PERSONNEL

- A. O&M manuals shall be prepared and shall include, but not be limited to, the following categories:
1. Contractual and legal guides
 2. The purpose of the installation
 3. Installation records
 4. Description of the installation;
 5. Equipment schedule
 6. Parts identification and recommended spares
 7. Commissioning data;
 8. Operation and how installation is to be used and keep operational;
 9. Maintenance schedules;
 10. How installation may be modified or changed;
 11. Disposal of installation;
 12. Names and addresses of manufacturers;
 13. Set of "as built drawings"
 14. Emergency operating procedures
 15. Manufacturers literature;
- B. The services of a factory trained, field service representative shall be provided to supervise the final testing, charging, initial start-up and conduct concurrent operator instruction where applicable.

1.5 TRAINING

- A. Upon completion and acceptance of Installation, properly acquaint Representative of the ENGINEER with details and sufficient training to enable them to obtain maximum efficiency in the system operation.

1.6 DESIGN CRITERIA, REGULATIONS AND STANDARDS

- A. All valves are to be rated for the highest pressures and temperatures in the respective system in accordance with BS, but not less than 800 kPa (8.0 bar) unless specifically indicated otherwise.
- B. Where valves are specified for individual mechanical services (i.e. hot water heating, steam, etc.) all valves shall be of the same manufacturer unless prior written approval is obtained from ENGINEER.
- C. Reference Standards
1. BS 759 - Part1: Valves, mountings and fittings
 2. BS 1010 Draw off taps and stop valves for water services (screw down pattern)
 3. BS 1010 - Part2: Draw off taps and aboveground stop valves.
 4. BS 1212: Float operated valves (excluding floats).
 5. BS 1212 - Part1: Piston type
 6. BS 1212 - Part2: Diaphragm type (brass body)
 7. BS 1212 - Part3: Diaphragm type (plastics body)
 8. BS 1503: Steel forgings for pressure purposes.
 9. BS 1968: Floats for ball valves (copper).
 10. BS 2456: Floats (plastic) for ball valves for hot and coldwater.
 11. BS 2767: Manually operated copper alloy valves for radiators.
 12. BS 2879: Draining taps (screwdown pattern).



13. BS 5150: Cast iron wedge and double disc gate valve for general purpose.
14. BS 5151: Cast iron gate (parallel slide) valves for general purposes.
15. BS 5152: Cast iron globe and globe stop and check valves for general purposes.
16. BS 5153: Cast iron check valves for general purposes.
17. BS 5154: Copper alloy globe, globe stop and check, check and gate valves.
18. BS 5156: Diaphragm valves.
19. BS 5157: Steel gate (parallel slide) valves for general purposes.
20. BS 5158: Cast iron plug valves.
21. BS 5159: Cast iron and carbon steel ball valves for general purposes.
22. BS 5160: Steel globe valves, globe stop and check valves and lift type check valves.
23. BS 5163: Predominantly key-operated cast iron gate valves for water works purposes.
24. BS 5235: Dial-type expansion thermometers.
25. BS 5353: Steel plug valves.
26. BS 5433: Underground stop valves for water services.
27. BS 6683: Guide to installation and use of valves.
28. BS 6755: Testing of valves.
29. BS 6759: Safety valves.
30. BS 7350: Double regulating globe valves and flow measurement devices for heating and chilled water systems.
31. BS 7478: Guide to selection and use of thermostatic radiator valves.
32. BS EN 593: Industrial valves-metallic butterfly valves.
33. BS EN 837: Pressure gauges.
34. BS EN 837-1: Bourdon tube pressure gauges - dimensions, metrology, requirements and testing.
35. BS EN 1982: Copper and copper alloy-ingots and castings.
36. BS EN 10213: Technical delivery conditions for steel castings for pressure purposes.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Generally, all valves of the same type shall be of the same manufacturer. All gate, globe, angle, and swing check valves as a group shall be of the same manufacturer.
- B. All valves 50 mm and smaller shall be threaded and have bronze bodies.
- C. All valves 65 mm and larger shall be Iron Body Bronze Mounted (IBBM) type, i.e. with bronze trim, and shall be flanged (or grooved for grooved coupling joints).
- D. All valves 100 mm and larger mounted in excess of 2.15 m above the floor in mechanical rooms shall be equipped with chain operators. Extend chains to within 2 m of floor.
- E. Each valve shall be marked (engraved, stamped, or cast on each valve or metal tag, permanently attached to the valve) at the factory with the following minimum information:
 1. Manufacturer's Name.
 2. Catalogue or Figure No.
 3. Size and Pressure Class.
- F. Arrows to indicate direction of flow on check, globe, angle, non-return, and eccentric plug valves.

2.2 GATE VALVES

- A. General



1. Furnish valves designed for minimum PN16 (or PN20 as specified) non shock water, oil and gas working pressure.
 - B. Size 50 mm and Smaller
 1. Furnish bronze valves with screwed-in bonnet, non-rising stem, solid wedge disc, and threaded ends.
 - C. Size 65 mm and Larger
 1. Furnish Iron Body Bronze Mounted (IBBM) valves, i.e. cast iron body bronze trim valves, with bolted bonnet, non-rising stem, solid wedge disc, flanged ends, and renewable seat rings.
- 2.3 GLOBE VALVES
- A. Size 50mm and Smaller
 1. Furnish valves designed for minimum PN16 nonshock water, oil, and gas working pressures. Valves shall have union bonnets, integral seats, and renewable teflon discs. Valves shall permit disc and bonnet replacement without removing valves from piping.
 - B. Size 65 mm and Larger
 1. Furnish valves designed for minimum PN16 water, oil, and gas working pressure. Valves shall have bolted bonnets, guided bronze or bronze faced disc, outside screw and yoke, (O S and Y) and flanged ends. Discs and seats shall be renewable without removing valves from line.
- 2.4 BALANCING VALVES
- A. Balancing valves upto size 50mm dia shall be 16 bar rated, bronze variable orifice double regulating valve with threaded ends, screwed bonnet, parabolic slotted regulating disk, double regulating device, pressure test points.
 - B. Balancing valves size 65mm dia and above shall be 16 bar rated, cast iron variable orifice double regulating valves with flanged ends, inside screw, no rising stem , EPDM coated regulating disk, double regulating device with indicator , locking device and pressure test points.
- 2.5 CHECK VALVES
- A. Size 50 mm and smaller
 1. Furnish swing valves designed for minimum PN16 nonshock water working pressures. Valves shall have renewable discs and side plugs and regrindable, integral seats. Discs shall be renewable and seats regrindable without removing valves from the line.
 - B. Size 65 mm and Larger
 1. Valves shall be silent type spring loaded of the double door or wafer style. Valves shall be designed for minimum PN16 non shock water working pressure.



2. Double door valves shall have cast iron body, aluminum bronze or stainless steel disc, stainless steel spring and hinge pin, and Butyl rubber sheet. Wafer style valves shall have cast iron or semi-steel body, bronze disc, stainless steel spring, and Butyl rubber sheet.

2.6 PRESSURE REDUCING VALVES

A. Domestic Water Lines.

1. Pressure reducing valves shall be pilot controlled, hydraulically operated, diaphragm type with a low by-pass capability. The low-flow by-pass capability shall be achieved by using a balanced direct acting PRV as an integral part of the main valve. At very low flows when the main valve is almost completely closed, to prevent the possibility of cavitation the direct acting valve shall by-pass the main valve and maintain flow.
2. PRVs shall be bronze construction including the trim. The pressure reducing valves shall be suitable for maximum working pressure that exist within the system and downstream pressure should be site adjustable between 2 and 4 bar. Refer to Schematic drawings for the minimum locations at which PRV's shall be required.

2.7 RELIEF VALVES

A. Domestic Water Temperature and Pressure Relief Valve.

1. On hot water storage tanks provide an ASME rated thermostatic, self-closing, temperature and pressure relief valve, located in the relief valve openings of tanks. Valve shall have a minimum thermal discharge capacity equal to the input capacity of the heater standard pressure setting of 600 kPa and standard temperature setting of 100 degrees C. Relief valve pipe to discharge to floor drain.

2.8 BALL VALVES

A. General

1. Valves shall be non-blowout stem design. Quarter turn of handle shall fully open or close valve. Handle position shall indicate whether valve is open or closed. Handle stops shall be a permanent, integral part of the body.

B. Size 50 mm and Smaller

1. Valves shall be standard port, 2-piece construction with screwed ends. Valves shall be designed for minimum PN25.
2. Valves shall have bronze or brass body, chromium plated or stainless steel ball, steel handle with vinyl grip, and replaceable teflon seats.

C. Size 65 mm and Larger

1. Valves shall be standard port, BS 5159 with flanged ends. Valves shall be designed for minimum PN16 working pressure.
2. Valves shall have steel body, chrome or nickel plated steel or stainless steel ball, replaceable teflon seats, and steel stem and handle.
3. Where required for installation in the fire protection system, the ball valves shall be FM approved.



2.9 FLOAT VALVES

- A. Float valves shall be installed as indicated in the drawings to provide consistent level control in reserve supply water storage tanks. The valve shall meet the requirements of the Water Byelaws for air gaps and shall be constructed throughout in approved materials and shall prevent back siphoning. Inlet flow shall be co-axial with the piston movement, to ensure high discharge capacity and smooth, quiet operation. The valve shall have 'O' ring piston seals, resilient seated disk, Dezincification resistant bronze fabricated approved solder and shall be hydraulically tested to ensure buoyancy and constructed in accordance with BS1968.
- B. Bronze equilibrium float valves 80 and above shall be flanged end, flat faced and drilled to suit BS4504 PN16.
- C. Bronze equilibrium float valves upto 50 shall be screwed end BS2779 parallel and shall be provided complete with back nut.
- D. Floats for valve sizes 80mm and above shall be of copper.

2.10 SOLENOID VALVES

- A. Electrically operated solenoid valves shall be single phase 240V and shall be rated for the system pressure.

2.11 BIB TAPS

- A. Bib-cocks shall be in accordance with BS 1010 : 1973. They shall be provided with hose union nosepiece and hand wheel operated.

2.12 AUTOMATIC AIR VALVES

- A. Automatic air valves shall have a bronze body with bolted cover and a 9mm top outlet.
- B. They shall each incorporate a suitable float mounted on a stainless steel spindle terminating in a needle valve for closure against a stainless steel seating at the outlet. For low pressure systems the valve shall be provided with a test cock and a brazed float, but for medium and high pressure systems the float shall be of stainless steel.
- C. Automatic air valves for low pressure systems shall, incorporate an internal ball check valve at inlet to prevent air entry to the system. Automatic air valves for cold water services shall be to the approval of the local water authority.

2.13 STRAINERS

- A. Up to and including DN50 strainers shall be manufactured from bronze and shall be of the 'Y' type with bolted cap, PN25 temperature/pressure rating. Strainers above DN50 shall be manufactured from cast iron and shall be of the 'Y' type with bolted cap, PN16 temperature/pressure rating and shall be complete with drilled and tapped caps complete with drain cocks. Strainers shall be provided with a medium grade screen sized such that in their clean condition the maximum pressure drop at the design flow rate shall not exceed 6kPa. For chilled water the mesh shall be stainless steel and for water services the mesh shall be of copper.



2.14 SUCTION DIFFUSER

- A. Install suction diffuser where specified and as shown on drawings.
1. The suction diffuser shall be flanged type, suitable for a maximum working pressure of 16 bar. The flow straightening fitting shall be of cast iron flanged type for connection to pump. The fitting shall have a stainless steel combination diffuser – strainer-orifice cylinder with 3/16" perforations to protect the system pump. The full length stainless steel flow straightening vanes shall provide non-turbulent flow to the suction side of the pump. The startup strainer shall be of 16mesh bronze and support foot shall be provided to eliminate the pipe strain at the floor fitting / pump connection. All Internal components shall be replaceable type.

2.15 MULTIFUNCTION THERMOSTATIC CIRCULATON VALVE

Domestic hot water installations should be balanced with thermostatic balancing valve for a constant water temperature in the system with following characteristics:

- Temperature setting range: 35 - 60 °C
- Temperature pre-setting values visible on a visual scale
- Setting should be hidden or lockable to prevent unauthorized change (Allen key)
- Thermostatic module should be integrated to ensure accurate temperature control
- Optional upgrade with thermal disinfection module to prevent Legionella bacteria in hot water
- systems: - automatic (self-acting) disinfection or electronically controlled disinfection (controller
- with thermal actuator and temperature sensors) Disinfection upgrade can be done during operation, under pressure
- Thermo element should be replaceable for easier
- Material in contact with water should be compliant with No Lead Brass EU directive
- Valve body should be made of Bronze for better durability and corrosion resistance
- Water temperature measurement should be optional
- Valve should have flow indication mark
- Reliable packaging for safe transport and handling

Product characteristics

- Pressure class: PN 10
- Maximum medium temperature: 90°C[MP3]
- Connection size: DN 15-20
- Connection type: Internal thread ISO 7/1
- Maximum differential pressure: 100 kPa
- Factory pre-setting: 50 °C
- Installation: valve should be mounted in the circulation pipe
- Shall be WRAS , DVGW or KIWA approved .

PART 3 - EXECUTION

3.1 PREPARATION

- A. Provide valves at each piece of equipment to isolate equipment from its connected system. Locate strainers and valves as necessary to provide easy isolation and cleaning of strainers.



- B. Strainers shall be installed ahead of all automatic valves and elsewhere as indicated on Drawings. Provide a ball valve and 1.5 m of rubber hose in the blow off opening of each strainer.
- C. Unions shall be provided adjacent to each screwed type valve and shall be on the outlet side of the valve.

3.2 INSTALLATION/APPLICATION/PERFORMANCE/ERECTION

- A. Properly align piping before installation of valves in an upright position; operators installed below the valves will not be accepted.
- B. Install valves in strict accordance with valve manufacturer's installation recommendations. Do not support weight of piping system on valve ends.
- C. Install all valves with the stem in the upright position. Valves may be installed with the stem in the horizontal position only where space limitations do not allow installation in an upright position or where large valves are provided with chain wheel operators. Where valves 65mm and larger are located more than 4m above mechanical room floors, install valve with stem in the horizontal position and provide a chain wheel operator. Valves installed with the stems down, will not be accepted.
- D. Prior to flushing of piping systems, place all valves in the full-open position.
- E. Flanged valves shall be installed between flanges.
- F. Shut-Off Valves
 - 1. Install shut-off valves at all equipment, at each branch take-off from mains, and at each automatic valve for isolation or repair.
- G. Balancing Valves
 - 1. Provide balancing valves for all major equipment and at each major branch takeoff and at the discharge of each pump as indicated on drawings and details.
- H. Calibrated balance valves:
 - 1. Install where indicated on the drawings and details for balancing of hydronic systems.
- I. Drain Valves
 - 1. Provide drain valves for complete drainage of all systems. Locations of drain valves include low points of piping systems, equipment locations specified or other locations required for drainage of systems.
- J. Safety Relief Valves
 - 1. Use air pressure to clean piping prior to installation of safety relief valves.
 - 2. Install relief valves in locations indicated on drawings, downstream of all pressure reducing valves.
 - 3. Install valves in the vertical position, with drain holes, including those from dip pan elbows, piped to the nearest drain. Inlet and outlet piping connecting to valves must be the same size as valve
 - 4. Install drip pan as detailed at first vertical rise of the vent pipe. Keep pipe between safety valve and drip pan as short and straight as possible. Support piping and drip pan elbow independently to prevent stress at connections to safety valves. Install



vent pipe so that its weight does not rest on the drip pan elbow. Extend drain line from drip pan elbow and relief valve to nearest drain. Pipe discharge from water system relief valves to nearest drain.

K. Spring Loaded Check Valves

1. Install a spring loaded check valve in each pump discharge line where two pumps operate in parallel and no combination shutoff, check and balancing valve is being used.

L. Swing Check Valves

1. Provide swing check valves where specified, detailed. In such cases, provide isolation valves to allow repair or replacement of check valve.

M. Combination Shut-Off, Check, And Balancing Valves

1. Contractor may use combination shut-off, check and balancing valves where separate shut-off valve, check valve, and balancing valve are specified or detailed in pump discharge piping.

N. Pressure Reducing Valves

1. Provide gate valve and strainer at inlet. Provide gate valve at outlet. Install pressure gauges to indicate inlet and outlet pressure at each pressure reducing valve.
2. Use concentric reducers at inlet and outlet of reducing valves where connections are not the same size as adjacent piping.

3.3 ADJUSTMENT AND CLEANING

- A. Valves and stops shall be adjusted, packed, and repacked as may be required to eliminate leaks and produce proper flow.

END OF SECTION 22 05 23